2020年12月长篇阅读解析第1套

人工智能在农业领域所面临的挑战

一、总体分析

本文选自 *AgFunderNews*（农业新闻网站）2017 年 2 月 20 日文章 The Challenges for Artificial Intelligence in Agriculture(人工智能在农业领域所面临的挑战)。文章围绕人工智能在农业领域所面临的挑战展开，先指出人工智能在农业领域的发展并不顺利，随后分析原因，最后肯定人工智能的潜能，展望其在农业领域的发展前景。

二、试题分析

试题	定位词	核心义
36. Farmers will not profit from **replanting**① once they have **applied**② most of the **fertilizer**③ and other chemicals to their fields.	[1] not profit from replanting [2] most of the fertilizer and other chemicals	种植户补种时间要求：在施用大部分肥料和其他化学品之前完成。
37. Agriculture differs from the medical science of the human body in that its environment is not a **contained**④ one.	[1] human body [2] not a contained one	农业与人体医学的区别：前者并非受控制的环境。

续表

试题	定位词	核心义
38. The agronomist is sure that he will obtain a near accurate count of plant **population**⑤ with his software.	[1] The agronomist [2] sure [2] a near accurate count	农学家确信之事：其软件可以生成近乎精确的植株数量。
39. The application of artificial intelligence to agriculture is much more challenging than to most other industries.	[1] much more challenging [2] most other industries	农业领域人工智能应用的难度：远大于大多数其他行业。
40. Even the farmers know the data provided by the UAV is not correct.	[1] the data provided by the UAV [2] not correct	无人机提供的数据：明显有误。
41. The pressure for quick results leads to product failure, which, in turn, arouses doubts about the applicability of AI technology to agriculture.	[1] The pressure for quick results [2] product failure [3] arouses doubts	要求速见成效的后果：产品失败，人工智能技术遭质疑。
42. **Remote sensors**⑥ are aimed to help farmers improve decision-making to increase **yields**⑦.	[1] Remote sensors [2] improve decision-making to increase yields	遥感器旨在达到的效果：帮助种植户改善决策，提高产量。
43. The farmer expects the software to tell him whether he will have to replant any parts of his farm fields.	[1] the software [2] replant any parts of his farm fields	种植户对专业软件效果的预期：帮助确定是否需要补种。
44. Agriculture proves very difficult to **quantify**⑧ because of the constantly changing conditions involved.	[1] difficult to quantify [2] constantly changing conditions	农业难以量化的原因：涉及的条件时刻在变。
45. The same seed and fertilizer program may yield completely different outcomes in different places.	[1] The same seed and fertilizer program	同一播种和施肥计划在不同地区的效果：可能截然不同。

·词汇注释与难句分析·

①**replant** [ˈriːˈplɑːnt] *v.* 再栽；补种
②**apply** [əˈplaɪ] *v.* 敷用；施用
③**fertilizer** [ˈfɜːtəlaɪzə(r)] *n.* 肥料
④**contained** [kənˈteɪnd] *a.* 受控制的
⑤**population** [ˌpɒpjuˈleɪʃn] *n.* 总数
⑥**remote sensor** 遥感器
⑦**yield** [jiːld] *n.* 产量
⑧**quantify** [ˈkwɒntɪfaɪ] *v.* 以数量表述；量化

定位词选取原则：(1)排除多次复现的主题性信息，如 Farmers、Agriculture、artificial intelligence/AI technology 等；(2)优先选择人、物等名词(短语)，如 The agronomist、human body、Remote sensors、the software、The same seed and fertilizer program 等；(3)补充选取其他具有明显区辨性的细节信息，如 most of the fertilizer and other chemicals、a near accurate count、most other industries 等含有数量修饰成分的名词短语，sure、much more challenging、not correct 等形容词(短语)，not profit from replanting、improve decision-making to increase yields 等动词短语。

解题思维原则：速读巧解时，带着试题定位词快速浏览文中段落，一旦发现试题定位词(不是单一、而是全部)出现于某段某处，随即将试题核心义与该定位处核心文意进行比对，若比对成功即可确定答案，正确答案往往是对原文定位处的同义复现或改写，偶尔可能是几个句子的文意概括或段落主旨概括。(短时记忆有限者，这一解题过程可分批进行，如先分析 5 道题并通读全文，然后分析剩下 5 道题及剩余未确定段落。)

三、速读巧解 VX:kypk2023

The Challenges for Artificial Intelligence in Agriculture

A) ❶A group of corn farmers stands **huddled**① around an *agronomist*(农学家) and his computer on the side of an **irrigation**② machine in central South Africa. ❷The agronomist

人工智能在农业领域所面临的挑战

在南非中部的一台灌溉机旁，一群玉米种植户聚集在一位农学家和他的电脑周围。这位农学家刚刚使用一

has just flown over the field with a **hybrid unmanned aerial vehicle (UAV)**③ that takes off and lands using **propellers**④ yet maintains distance and speed for **scanning**⑤ vast **hectares**⑥ of land through the use of its **fixed wings**⑦.

B) The UAV is fitted with a four **spectral band**⑧ **precision sensor**⑨ that conducts **onboard**⑩ processing immediately after the flight, allowing farmers and field staff to address, almost immediately, any crop **abnormalities**⑪ that the sensor may have recorded, making the data collection truly **real-time**⑫.

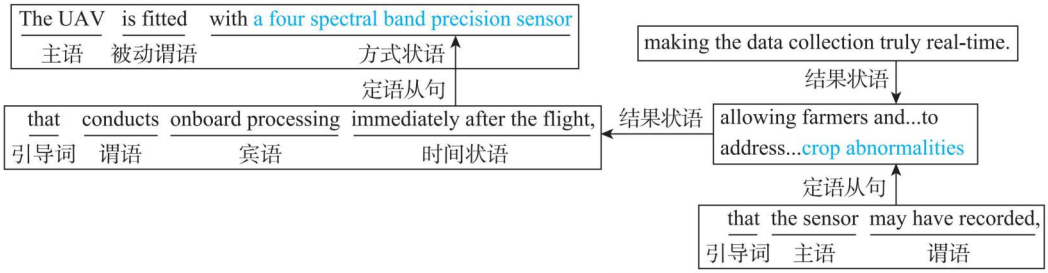
架混合式无人机(UAV)飞过周围田地,该无人机使用螺旋桨起飞及降落,借助固定翼保持距离及速度,来扫描大面积的田地。VX:kypk2023

这架无人机装有一个四光谱波段精密传感器,可在飞行结束后即刻进行机载处理,这让种植户及田间工作人员几乎立即便可处理传感器所记录到的任何作物异常状况,使数据采集真正实现了实时性。

·词汇注释与难句分析·

①**huddle** [ˈhʌdl] *v.* 挤作一团;聚在一起
②**irrigation** [ˌɪrɪˈgeɪʃn] *n.* 灌溉
③**hybrid unmanned aerial vehicle** 混合式无人机
④**propeller** [prəˈpelə(r)] *n.* 螺旋桨
⑤**scan** [skæn] *v.* 扫描;扫掠
⑥**hectare** [ˈhekteə(r)] *n.* 公顷
⑦**fixed wing** 固定翼
⑧**spectral band** 光谱段
⑨**precision sensor** 精密传感器
⑩**onboard** [ˈɒnˈbɔːd] *a.* 机载的
⑪**abnormality** [ˌæbnɔːˈmæləti] *n.* 不正常,反常
⑫**real-time** [ˈriːlˈtaɪm] *a.* 实时的

The UAV is fitted with a four spectral band precision sensor that conducts onboard processing immediately after the flight, allowing farmers and field staff to address, almost immediately, any crop abnormalities that the sensor may have recorded, making the data collection truly real-time.



·速读巧解·

A—F 段以一次失败的试验为例,说明人工智能在农业领域的应用现状:尚无法实现精确预测。

A、B 段介绍试验的基本信息。

A 段引出试验场景及试验工具(UAV)。开篇以场景"一群玉米种植户聚集在农学家周围"引入一次农田技术试验,并明确试验地点"南非中部某处玉米地";随后引出农学家使用的技术工具 UAV,并说明其运行细节及基本功用"扫描大面积农田"。**全段核心信息:**(1)试验地点;(2)UAV 的功用。

【定位词定位】无。

B 段具体介绍试验所用技术"UAV 机载精密传感器"。首先以 The UAV is fitted with 引出机载设备"四光谱波段精密传感器";随后 that conducts...allowing...making... 串联起对该传感器的介绍:工作方式(即刻进行机载处理)→作用(几乎立即便可处理任何作物异常情况)→意义(数据采集真正实现了实时性)。immediately、almost immediately、real-time 凸显其"及时性/实时性"特点。**全段核心信息:**UAV 机载精密传感器的作用机制。

【定位词定位】无。

C) ❶In this instance, the farmers and agronomist are looking to **specialized**① software to give them an accurate plant population count. ❷It's been 10 days since the corn emerged and the farmer wants to determine if there are any parts of the

在此情况下,种植户和农学家希望专业软件为其提供准确的植株数量。玉米出苗已有 10 天,种植户想确定是否有田块因出苗不足或风害倒

field that require replanting due to a lack of emergence or **wind damage**②, which can be severe in the early stages of the summer rainy season.

D) ❶At this growth stage of the plant's development, the farmer has another 10 days to conduct any replanting before the majority of his fertilizer and chemical applications need to occur. ❷Once these have been applied, it becomes economically **unviable**③ to take corrective action, making any further collected data historical and useful only to inform future practices for the season to come.

伏而需补种，这些问题在夏天雨季的初期会很严重。

在该作物发育的这一生长阶段，种植户还有 10 天的时间进行补种，之后才需要施用大部分肥料和化学品。然而一旦施用了这些肥料和化学品，再采取补救措施在经济上便不可行了，这使得进一步收集的任何数据都已过时，只能为下一季的田间活动提供参考。

VX:kypk2023

·词汇注释与难句分析·

①**specialized** ['speʃəlaɪzd] *a.* 专业的；专门的
②**wind damage** 风灾；风害
③**unviable** [ʌn'vaɪəbl] *a.* 不可行的

·速读巧解·

C、D 段介绍试验的预期效果：精准、快速统计植株数量，以及时确定补种计划。

C 段明确实验的预期效果，突出"精确性"，暗示"时效性"。In this instance、specialized software 承接上文 UAV 机载精密传感器的一般作用，聚焦本次试验的预期效果。... looking to... wants to determine... 呈递"预期效果（利用专业软件准确计算植株数量）→使用目的（帮助农民确定是否有田块需要补种）"的逻辑。It's been 10 days since the corn emerged 暗示时效性"玉米已经出苗 10 天，补种正当时"。**全段核心信息：**种植户和农学家对软件效用的预期。

【定位词定位】43 题 the software 对应①句 specialized software，replant any parts of his farm fields 同义改写②句 there are any parts of the field that require replanting。

【核心义匹配】本段种植户对专业软件效用的期待"帮助其确定是否有田块需要补种"匹配 43 题核心义。

D 段补充玉米补种的相关知识，强调"时效性"。①句 At this growth stage、another 10 days 接续 C 段时间点"玉米出苗已有 10 天"，明确补种时间要求"未来 10 天内完成（another 10 days）"，即植株统计必须尽快。②句以 Once 假定情形，说明错过补种期的后果"大部分肥料和化学品已经施用，补种失去经济效益"，强调数据搜集的时效性。**全段核心信息：**(1)玉米补种期；(2)错过补种期的后果；(3)数据时效性。

【定位词定位】36 题 not profit from replanting 同义改写②句 economically unviable to take corrective action，most of the fertilizer and other chemicals 复现①句 the majority of his fertilizer and chemical...。

【核心义匹配】②句种植户的补种时机"施用肥料和化学品后再补种在经济上没有效益"匹配 36 题核心义。

E) ❶The software completes its processing in under 15 minutes producing a plant population count map. ❷It's difficult to **grasp**① just how **impressive**② this is, without understanding that just over a year ago it would have taken three to five days to process the exact same data set, **illustrating**③ the advancements that have been achieved in **precision agriculture**④ and remote sensing in recent years. ❸With the software having been developed in the United States on the same variety of crops in seemingly similar conditions, the agronomist feels confident that the software will produce a near accurate result.

该软件在 15 分钟内即可完成数据处理，生成植株数量图。倘若不了解仅仅在一年多以前，处理完全相同的数据集还需要三到五天，就很难理解这有多么令人赞叹，这展现了近些年来精准农业和遥感技术所取得的进步。由于该软件是美国针对看似相同条件下的同品种作物开发出来的，所以农学家信心十足地认为它会得到近乎准确的结果。

F) ❶As the map appears on the screen, the agronomist's face begins to drop. ❷Having walked through the planted **rows**⑤ before the flight to gain a **physical**⑥ understanding of the situation on the ground, he knows the instant he sees the data on his screen that the plant count is not correct, and so do the farmers, even with their limited understanding of how to read remote sensing maps.

当植株数量图出现在屏幕上时，农学家的脸色沉了下来。无人机起飞前，他曾走过作物行间，对田间情况做了实地了解，当他从屏幕上看到数据的那一刻，他就立马知道植株数量不对；种植户也发现不对，虽然他们对如何解读遥感地图了解有限。

· 词汇注释与难句分析 ·

①**grasp** [grɑːsp] *v.* 理解；领会
②**impressive** [ɪmˈpresɪv] *a.* 令人赞叹的
③**illustrate** [ˈɪləstreɪt] *v.* 说明；解释
④**precision agriculture** 精准农业
⑤**row** [rəʊ] *n.* 一行/排/列
⑥**physical** [ˈfɪzɪkl] *a.* 现实的；物质的

· 速读巧解 ·

E、F 段介绍试验结果：软件计算速度快但结果不准确。

E 段指出软件计算速度非常快，农学家们对试验结果相当自信。①②句先明确试验中软件的计算速度“15 分钟内完成(in under 15 minutes)”，后以 impressive(令人赞叹的)高度评价其速度表现，并对比一年前所需时间(3 到 5 天)凸显其进步之飞速，最后以 illustrating 引出其意义：展现了近些年来在精准农业和遥感方面的巨大进步。③句主句以 confident... near accurate 明确农学家的乐观预期“将得出近乎精确的结果”；With 引导的状语说明原因：软件开发时使用的作物品种相同，实验条件极其相似(on the same variety of crops in seemingly similar conditions)。**全段核心信息：**(1)软件的速度表现；(2)农学家对试验结果的乐观预期。

【定位词定位】38 题 The agronomist 复现③句原词，sure、a near accurate count 同义复现③句 confident、a near accurate result。

【核心义匹配】③句农学家对其软件的态度“信心十足地认为它会得出近乎精确的结果”匹配 38 题核心义。

F 段指出随后生成的结果明显不准确。①句以农学家看到数据后的神情变化(face begins to drop)暗示试验结果不如人意。②句 is not correct 明示试验结果“计算不准确”；he knows the instant... and so do the farmers, even with... limited... 则以“不仅农学家一眼看出数量不对，就连对软件缺乏了解的种植户也是如此”强调计算结果精确性严重不足。**全段核心信息：**(1)软件的精确性表现；(2)农学家、种植户对试验结果的判断。

【定位词定位】40 题 the data provided by the UAV 对应②句 the data，not correct 在②句中复现。

【核心义匹配】②句信息“种植户也知道软件得出的数据明显不正确”匹配 40 题核心义。

G) **Hypothetically**①, it is possible for machines to learn to solve any problem on earth relating to the physical **interaction**② of all things within a **defined**③ or contained environment by using artificial intelligence and **machine learning**④.

H) ❶Remote sensors enable *algorithms*(算法)to **interpret**⑤ a field's environment as statistical data that can be understood and useful to farmers for decision-making. ❷Algorithms process the data, adapting and learning based on the data received. ❸The more **inputs**⑥ and statistical information collected, the better the algorithm will be at predicting **a range of**⑦ outcomes. ❹And the aim is that farmers can use this artificial intelligence to achieve their goal of a better **harvest**⑧ through making better decisions in the field.

假定来看，通过使用人工智能和机器学习，机器学会解决地球上任何关于“确定或受控制的环境中万物之间的物理交互”的问题是完全可能的。

遥感器使算法能将田间环境翻译为统计数据，种植户们可以理解这些数据并借其做出决策。算法对数据进行处理，基于收到的数据进行调整和学习。收集的输入和统计信息越多，算法预测一系列结果的能力就越强。这样做的目的是让种植户能够利用这种人工智能，在田间做出更好的决策，从而达到增加收成的目的。

· 词汇注释与难句分析 ·

①**hypothetically** [ˌhaɪpəˈθetɪklɪ] *ad.* 假设地
②**interaction** [ˌɪntərˈækʃn] *n.* 交互；互动
③**defined** [dɪˈfaɪnd] *a.* 明确的；清晰的
④**machine learning** 机器学习
⑤**interpret** [ɪnˈtɜːprət] *v.* 诠释；说明
⑥**input** [ˈɪnpʊt] *n.* 输入；输入的信息
⑦**a range of** 一系列
⑧**harvest** [ˈhɑːvɪst] *n.* 收成；收获

· 速读巧解 ·

G—J 段从上文的实例介绍转入理论分析，说明理论上人工智能在农业领域的巨大潜能。

G、H 段分析人工智能在农业领域的潜能。

G 段概述人工智能的巨大潜能和作用机制。Hypothetically，it is possible 引出关于“人工智能(machines/artificial intelligence/machine learning)潜能”的理论假定/理想情形，绝对化表达 any problem、all things 凸显人工智能的巨大潜能：解决地球上确定或受控制的环境中，有关万物之间物理交互的任何问题。**全段核心信息：**人工智能的潜能和作用机制。

【定位词定位】无。

H段具体介绍人工智能在农业领域的潜能。Remote sensors 呼应上文 precision sensor、remote sensing，说明农业领域所借助的主要技术：遥感技术。段内以 Remote sensors enable algorithms... Algorithms process the data... And the aim is that... 明确遥感器技术的作用机制：利用算法得到田间数据→对数据进行处理，基于收到的数据进行调整和学习→帮助种植户及时做出更好的决策→收成得以提高。**全段核心信息：**遥感器在农业领域的巨大潜能及作用机制。

【定位词定位】42 题 Remote sensors 复现于本段段首，improve decision-making to increase yields 同义改写④句 achieve their goal of a better harvest through making better decisions in the field。

【核心义匹配】④句大意“使用遥感器是为了让种植户做出更好的决策，增加收成”匹配 42 题核心义。

I) ❶In 2011, IBM, through its R&D **Headquarters**① in Haifa, Israel, **launched**② an agricultural cloud-computing project. ❷The project, in **collaboration**③ with a number of specialized IT and agricultural partners, had one goal in mind—to take a variety of **academic**④ and physical data sources from an agricultural environment and turn these into **automatic**⑤ predictive solutions for farmers that would assist them in making real-time decisions in the field.

J) ❶Interviews with some of the IBM project team members at the time revealed that the team believed it was entirely possible to “algorithm” agriculture, meaning that algorithms could solve any problem in the world. ❷Earlier that year, IBM's **cognitive**⑥ learning system, Watson, competed in the game Jeopardy against former winners Brad Rutter and Ken Jennings with **astonishing**⑦ results. ❸Several years later, Watson went on to produce **ground-breaking**⑧ achievements in the field of medicine.

2011 年，IBM 通过其位于以色列海法的研发总部启动了一个农业云计算项目。该项目与 IT 及农业领域的许多专业伙伴进行了合作，时刻记住目标只有一个——从农业环境中获取各种纯理论的及物理数据源，并将其转化为自动预测解决方案，帮助种植户在田间做出实时决策。

当年对 IBM 项目组部分成员的采访表明，他们相信农业“算法化”是完全可行的，也即算法可以解决世界上任何问题。同年早些时候，IBM 认知学习系统“沃森”曾在（智力竞赛节目）《危险边缘》中对战前冠军布拉德·鲁特和肯·詹宁斯，战果惊人。若干年后，“沃森”又在医药领域取得了突破性的成就。

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·词汇注释与难句分析·

①**headquarters** [ˌhedˈkwɔːtəz] *n.* 总部
②**launch** [lɔːntʃ] *v.* 开展；发起
③**collaboration** [kəˌlæbəˈreɪʃn] *n.* 合作；协作
④**academic** [ˌækəˈdemɪk] *a.* 学术的；纯理论的
⑤**automatic** [ˌɔːtəˈmætɪk] *a.* 自动的
⑥**cognitive** [ˈkɒgnətɪv] *a.* 认知的
⑦**astonishing** [əˈstɒnɪʃɪŋ] *a.* 令人惊讶的；非常惊人的
⑧**ground-breaking** [ˈgraʊndˌbreɪkɪŋ] *a.* 开拓性的；突破性的

·速读巧解·

I、J段介绍 IBM 的农业云计算项目，说明人工智能的应用现状。

I段介绍 IBM 的农业云计算项目。①句引出该项目。②句以 had one goal in mind 引出项目目标，to take... and turn... that would... 明确目标实现步骤：从农业环境中获取各种纯理论及物理数据源→将其转化为自动预测方案→帮助种植户于田间做出实时决策。**全段核心信息：**IBM 农业云计算项目及其目标。

【定位词定位】无。

J段介绍 IBM 项目在其他领域取得的成就。①句介绍 IBM 团队当初的想法“算法是万能的，农业算法化完全可行”，再现 G 段的理想状况，同时暗示想法可能过于乐观。②③句指出 IBM 旗下认知学习系统“沃森”在智力竞赛、医药等多领域都取得了惊人成就，说明算法确实“强大”，同时暗藏“唯独农业领域未能成功应用人工智能”之意，为下文探讨作铺垫。**全段核心信息：**(1)IBM 团队看好农业算法；(2)“沃森”的优秀表现。

【定位词定位】无。

K)❶So why did the project have such success in medicine but not agriculture? ❷Because it is one of the most difficult fields to contain for the purpose of statistical quantification. ❸Even within a single field, conditions are always changing from one **section**① to the next. ❹There's unpredictable weather, changes in soil quality, and the ever-present possibility that

那么，为何该项目在医药领域大获成功，却在农业领域受阻？因为农业是最难进行统计量化控制的领域之一。即便是在单单一片农田内，每个地段的环境条件也总是各不相同。既有变化莫测的天气、千差万别的土质，

pests② and diseases may pay a visit. ❺Growers may feel their prospects are good for an **upcoming**③ harvest, but until that day arrives, the outcome will always be uncertain.

L) ❶ By comparison, our bodies are a contained environment. ❷ Agriculture takes place in nature, among ecosystems of interacting organisms and activity, and crop production **takes place**④ within that ecosystem environment. ❸But these **ecosystems**⑤ are not contained. ❹They are subject to **climatic occurrences**⑥ such as weather systems, which impact upon **hemispheres**⑦ as a whole, and from **continent**⑧ to continent. ❺ Therefore, understanding how to manage an agricultural environment means **taking** literally many hundreds if not thousands of factors **into account**⑨.

M) ❶ What may occur with the same seed and fertilizer program in the United States' Midwest region is almost certainly unrelated to what may occur with the same seed and fertilizer program in Australia or South Africa. ❷A few factors that could impact on **variation**⑩ would typically include the measurement of rain per unit of a crop planted, soil type, patterns of **soil degradation**⑪, daylight hours, temperature and so forth.

N) So the problem with **deploying**⑫ machine learning and artificial intelligence in agriculture is not that scientists lack the capacity to develop programs and **protocols**⑬ to begin to address the biggest of growers' concerns; the problem is that in most cases, no two environments will be exactly alike, which makes the testing, **validation**⑭ and successful **rollout**⑮ of such technologies much more **laborious**⑯ than in most other industries.

又始终存在害虫、病害来袭的风险。种植者或许会觉得前景大好、丰收在即，但只要一日不收获，最终的收成就总是未知。

相较之下，我们的身体是一个受控制的环境。农业活动在大自然中进行，在有机体相互作用的动态生态系统中进行，作物生产发生于这一生态系统环境内。但这些生态系统是不受控制的。它们易受天气系统等气候现象的影响，这种影响会覆盖整个半球，并从一个大洲传至另一大洲。因此，了解如何管控农业环境就意味着（毫不夸张地）要将即使没有几千也有几百种因素考虑在内。

同样一项播种和施肥计划在美国中西部地区的效果几乎必定与其在澳大利亚或是南非的效果毫无关联。造成这种差异的典型因素可能包括：种植作物地区的单位面积降雨量、土壤类型、土壤退化模式、日照时长、温度等等。

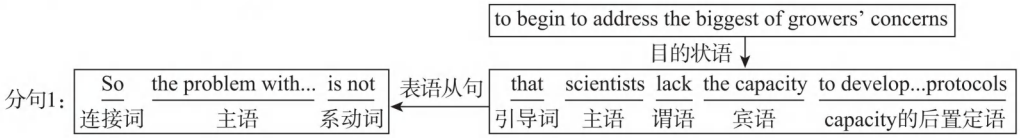
因此，农业领域运用机器学习和人工智能的问题并不在于科学家能力欠缺，不能开发出计划及协议来着手解决种植户最大的担忧；真正的问题在于，大多数情况下并不存在两个完全相同的环境，这使得此类技术的测试、验证和成功上市都比其他大多数行业更费劲。

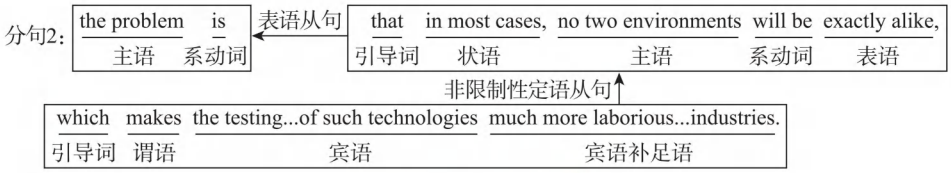
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· 词汇注释与难句分析 ·

①**section** [ˈsekʃn] *n.* 地段；平方英里（土地单位）
②**pest** [pest] *n.* 害虫
③**upcoming** [ˈʌpkʌmɪŋ] *a.* 即将来到的
④**take place** 发生
⑤**ecosystem** [ˈiːkəʊsɪstəm] *n.* 生态系统
⑥**climatic occurrence** 气候现象；气候事件
⑦**hemisphere** [ˈhemɪsfɪə(r)] *n.* 半球
⑧**continent** [ˈkɒntɪnənt] *n.* 大陆
⑨**take... into account** 考虑到；顾及
⑩**variation** [ˌveəriˈeɪʃn] *n.* 变化
⑪**soil degradation** 土壤退化
⑫**deploy** [dɪˈplɔɪ] *v.* 部署；调度
⑬**protocol** [ˈprəʊtəkɒl] *n.*（数据传递的）协议，规程
⑭**validation** [ˌvælɪˈdeɪʃn] *n.* 确证；验证
⑮**rollout** [ˈrəʊlaʊt] *n.* 正式推出
⑯**laborious** [ləˈbɔːriəs] *a.* 耗时费力的；辛苦的

So the problem with deploying machine learning and artificial intelligence in agriculture is not that scientists lack the capacity to develop programs and protocols to begin to address the biggest of growers' concerns; the problem is that in most cases, no two environments will be exactly alike, which makes the testing, validation and successful rollout of such technologies much more laborious than in most other industries.





·速读巧解·

K—N 段回应开篇事例，分析人工智能在农业领域折戟的原因：由农业自身特点所致。

K 段总领段群，概括原因：农业极难进行统计量化控制（极具不确定性）。①句 why 抛出问题：为何人工智能在医药领域大获成功，却在农业领域折戟？②句 Because 引出总体原因，it is one of the most difficult fields to contain 回应 G 段 within a defined or contained environment，明确原因。③至⑤句 always changing、unpredictable、changes、the ever-present possibility、always... uncertain 近义复现凸显农业的“不确定性”特点：农业生产环境变化莫测，难以收集可靠数据。**全段核心信息：**(1)IBM 项目在农业领域折戟；(2)农业环境不确定性强，难以统计量化。

【定位词定位】44 题 difficult to quantify 概括②句 one of the most difficult fields... for... statistical quantification，constantly changing conditions 改写③句 conditions are always changing。

【核心义匹配】②至⑤句内容“农业是最难统计量化的领域之一，因为其环境条件随时随地在变，不可预测”匹配 44 题核心义。

L 段通过对比人类身体，展现农业的不确定性：农业环境是不受控制的，诸多因素相互关联、牵一发而动全身。①至③句 By comparison、But 继续对比医药（our bodies 为医学研究对象，呼应 K 段 medicine）、农业两大领域，凸显后者“不受控制（not contained）”的特点。④⑤句具体说明该特点及影响：农业生产受到复杂外部环境的影响，其统计量化需考虑成百上千种因素，难度极大。**全段核心信息：**农业生产环境具有不受控制的特征，其统计量化涉及诸多因素。

【定位词定位】37 题 human body 对应①句 our bodies，not a contained one 复现③句 not contained。

【核心义匹配】①至③句大意“相较之下，人体是受控制的环境，而农业环境是不受控制的环境”匹配 37 题核心义。

M 段进一步展现农业的不确定性：地区差异大，同一计划在不同地区效果可能完全不同。①句以“同一播种和施肥计划可能在不同国家产生完全不同的效果”为例说明农业的地区差异之大；②句以 A few factors... would typically include... 补充列举造成该差异的若干典型原因。**全段核心信息：**同一农业计划在不同地区的影响可能完全不同。

【定位词定位】45 题 The same seed and fertilizer program 完整复现①句表达。

【核心义匹配】首句大意“同样的播种和施肥计划在美国西部、澳大利亚、南非等地的效果互不相干”匹配 45 题核心义。

N 段总结：农业领域人工智能应用的问题在于农业环境极度不可控。So 承上作结，the problem... is not that... the problem is that... 取舍结构重申原因：农业生产环境千差万别，农业领域人工智能技术的实测、验证和上市都远比其他行业更难。**全段核心信息：**人工智能在农业领域的应用远比其他行业更难。

【定位词定位】39 题 much more challenging 改写 much more laborious，most other industries 复现段末表达。

【核心义匹配】本段大意“农业领域的人工智能应用远比大多数其他行业更加费时耗力”匹配 39 题核心义。

O) ❶Practically, to say that AI and Machine Learning can be developed to solve all problems related to our physical environment is to basically say that we have a complete understanding of all aspects of the interaction of physical or material activity on the planet. ❷After all, it is only through our understanding of ‘the nature of things’ that protocols and processes are designed for the rational capabilities of cognitive systems to take place. ❸And, although AI and Machine Learning are teaching us many things about how to understand our environment, we are still far from being able to predict critical outcomes in fields like agriculture **purely**[①] through the cognitive ability of machines.

实际上，“发展人工智能和机器学习就能解决与物理环境相关的所有问题”这一想法本质上是认为我们能全面理解地球上物理或物质活动方方面面的相互作用。毕竟，只有凭借对“事物本质”的理解，我们才能设计出使认知系统产生理性能力的协议和流程。此外，虽然人工智能和机器学习在理解环境方面给我们以诸多教益，但仅凭机器的认知能力，我们还远远无法在类似农业这样的领域实现对关键结果的预测。

P) Backed by the **venture capital**② community, which is now investing billions of dollars in the sector, most agricultural technology **startups**③ today are pushed to complete development as quickly as possible and then encouraged to flood the market as quickly as possible with their products.

Q) ❶This usually results in a failure of a product, which leads to **skepticism**④ from the market and delivers a **blow**⑤ to the **integrity**⑥ of Machine Learning technology. ❷In most cases, the problem is not that the technology does not work, the problem is that industry has not taken the time to respect that agriculture is one of the most uncontained environments to manage. ❸For technology to truly make an impact on agriculture, more effort, skills, and funding is needed to test these technologies in farmers' fields.

在风投界(如今正将数十亿美元注入农业领域)的支持下,大多数农业技术初创企业都被催促着尽快完成开发,然后被怂恿着尽快推出产品挤占市场。

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这通常会导致产品失败,进而引发市场怀疑,并打击机器学习技术的信用。大多数情况下,问题并非是技术没有发挥作用,而在于业界未能花点时间来慎重对待"农业是最难约束、最难控制的环境之一"这一事实。若要技术真正对农业产生影响,我们需要投入更多努力、技术和投资在种植户的田地里对这些技术进行测试。

·词汇注释与难句分析·

①**purely** ['pjʊəli] *ad.* 仅仅;完全
②**venture capital** 风险资本
③**startup** ['stɑːtʌp] *n.* 初创企业;新兴公司
④**skepticism** ['skeptɪsɪzəm] *n.* 怀疑态度
⑤**blow** [bləʊ] *n.* 打击;挫折
⑥**integrity** [ɪn'tegrəti] *n.* 诚信

·速读巧解·

O—Q段提议:业界应尊重农业自身特点。

O段指出人类高估了自身对客观世界的认识,强调人工智能在农业领域的应用仍远未成熟。①句 Practically 回应 G 段 Hypothetically,表明本段开始从"假定理想情形"转入"分析现实情形"。整句以 to say... is to basically say... 展现等量关系:假定算法学习可以解决一切问题=认为人类已经完全掌握物质世界的规律。②句以 After all 引出原因解释。整句以 it is only through... that 强调必要条件关系:只有了解了事物本质,才能设计出具备理性能力的认知系统。③句以"让步—转折"结构 although... we are still far from being able to... 否定①②句暗留的可能,明确指出人工智能在农业领域的应用仍远未成熟。**全段核心信息:**人工智能应用的局限。

【定位词定位】无。

P段介绍农业技术的发展现状:大量资本涌入,一味追求速度。被动语态 are pushed、are... encouraged 连用,as quickly as possible 复现,共同凸显整个行业被资本裹挟、只求速度的不良现状。**全段核心信息:**农业技术行业的发展现状。

【定位词定位】41 题 pressure for quick results 是对 pushed to... as quickly as possible... then encouraged to... as quickly as possible 的概括,其他定位词无法匹配本段信息。

【核心义匹配】无。

Q段指出农业技术行业只求速见成效导致的不良后果,并就此提议。①句 results in、leads to、delivers a blow to 明示因果:一味追求速度→产品失败→市场怀疑→机器学习技术遭到质疑。②句为机器学习/人工智能技术正名,指出实际问题在于业内未能真正尊重农业自身的特点。③句 For... is needed 提议:应在技术的田间实测方面增加投入。**全段核心信息:**(1)农业技术行业一味追求速度的后果;(2)技术如何真正影响农业。

【定位词定位】37 题 not a contained one 复现②句 uncontained environments,human body 在本段无对应信息;41 题 The pressure for quick results 对应①句主语 This(回指 P 段 as quickly as possible 所凸显的行业乱象"大量资本涌入农技术行业,一味追求速度"),product failure 改写①句 a failure of a product,arouses doubts 改写 leads to skepticism。

【核心义匹配】①句大意"一味追求速度常导致产品失败,进而引发市场对农业领域机器学习技术应用的怀疑"匹配 41 题核心义。

R) ❶There is huge potential for artificial intelligence and machine learning to **revolutionize**① agriculture by integrating these technologies into critical markets on a global scale. ❷Only then can it **make a difference**② to the grower, where it really **counts**③. [1238 words]

人工智能与机器学习拥有巨大的潜力,可以通过将这些技术融入全球范围内的重要市场彻底变革农业。只有到那时,它才能为种植户带来改变,而这才是真正重要的。

·词汇注释与难句分析·

①**revolutionize** [ˌrevəˈluːʃənaɪz] *v.* 彻底改变；完全变革
②**make a difference** 起作用；有影响
③**count** [kaʊnt] *v.* 重要

·速读巧解·

R段总结全文，肯定人工智能的潜能，但指出其发展仍需时日且最终应惠及种植户。①句 huge potential... by... 明确肯定人工智能有望通过与全球重要市场整合而彻底革新农业。②句 Only then、really counts 表达对未来的希冀：农业人工智能真正惠及最需要帮助的种植户群体。**全段核心信息：**人工智能农业潜力巨大，有望惠及种植户。

【定位词定位】无。

四、答案精析

36. 种植户一旦在田地里施用了大部分肥料和其他化学品，就无法从补种中获利。

[答案] **D** [精解] D段①②句明确补种时机：一旦施用了大部分肥料和化学品，再进行补种在经济上便不可行了。②句中的 these 回指①句中的 the majority of his fertilizer and chemical...，take corrective action 呼应①句中的 conduct... replanting，试题是对该句信息的同义改写。

37. 农业与人体医学的区别在于其环境是不受控制的。

[答案] **L** [精解] L段前三句对比医药与农业，强调后者环境的特点"不受控制的"。试题是对二者区别的概括，differs from 体现了①句 By comparison、③句 But 所体现的比较之意；medical science 明确了①句 our bodies(呼应上段 medicine)所暗指的对象。

38. 农学家确信可以使用其软件获得一个近乎精确的植株数量。

[答案] **E** [精解] E段描述专业软件的性能，其中③句表明农学家对其软件性能的信心：(由于软件开发时所使用的作物品种及实验条件均相同/似)该软件将会得出近乎精确的结果。试题是对该句信息的同义改写，a near accurate count of plant population 对应 a near accurate result(result 指代该段首句 a plant population count map)。

39. 在农业领域应用人工智能比大多数其他行业更具挑战性。

[答案] **N** [精解] N段明确在农业领域应用人工智能的问题所在：人工智能技术的实测、验证和上市都远比其他大多数行业耗时费力。试题是对该内容的概括，application 对应 deploying。

40. 就连种植户都知道无人机提供的数据不正确。

[答案] **F** [精解] F段②句揭示专业软件所得结果：一看到屏幕上的数据，农学家立马知道植株数量不对；种植户也发现不对，虽然他们对如何解读遥感地图了解有限。试题是对该句信息的整合概括，Even 对应原文 even with their limited understanding of how to read remote sensing maps，强调"计算结果错误过于明显，连身为非专业人士的种植户都能看出来"。

41. 要求速见成效会导致产品失败，而产品失败又转而引发人们对人工智能技术在农业领域的适用性的怀疑。

[答案] **Q** [精解] Q段①句顺承上文内容指出一味追求速见成效的后果：产品失败，市场对机器学习技术在农业领域的适用性产生怀疑。试题是对该句的同义改写，句中 This 回指上段所述"一味求快"的行业乱象，The pressure for quick results 是对此内容的概括。

42. 使用遥感器的目的是帮助种植户改善决策，提高产量。

[答案] **H** [精解] H段介绍遥感器在农业领域的潜能，其中④句指出使用遥感器的目的/作用：让种植户能够利用这种人工智能在田间做出更好的决策，从而达到增加收成的目的。试题是对该句信息的整合，are aimed to help farmers... 同义改写 the aim is that farmers can...。

43. 种植户希望软件能告诉他农田里是否有任何需要补种的部分。

[答案] **C** [精解] C段说明种植户对专业软件效用的期待，①句提出期待"专业软件提供准确的植株数量"，②句解释说明期待"由此确定是否有田块需补种"。试题是对两句信息的提炼，expects 对应 are looking to，tell him whether... 对应 determine if...。

44. 事实证明，由于涉及的条件时刻都在变化，农业极难量化。

[答案] **K** [精解] K段②句指出农业是最难统计量化的领域之一，③句以"即使是在同一片农田内部，每个地段的环境条件都在时刻发生变化"凸显农业环境"变化莫测"的特点，说明其量化难度之大。试题是对两句大意的概括，because of 是对两句间暗含因果关联的明确。

45. 同样一项播种和施肥计划在不同地区可能产生截然不同的效果。

[答案] **M** [精解] M段①句指出一项播种和施肥计划在美国中西部的效果与其在澳大利亚、南非等地的效果互不相干，即同样的计划可能在不同地区产生完全不同的效果。试题是对该句大意的概括，completely different 明确 unrelated 的内涵，outcomes 改写 What may occur。